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A Raspberry Pi–based Intelligent System for Detecting Vitamin Deficiencies using Image Analysis and Lightweight Deep Learning Models

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Abstract - Vitamin deficiencies contribute to millions of preventable health complications worldwide, ranging from vision loss and neurological dysfunction to reduced immunity and skeletal weakness. In low-resource communities, laboratory testing is often inaccessible or costly, resulting in undiagnosed and untreated deficiencies. Many deficiencies, however, leave identifiable visual traces on anatomical regions such as the eyes, tongue, nails, and skin. This research presents a fully offline, low-cost, and portable diagnostic device built using Raspberry Pi, Pi Camera, TensorFlow Lite deep learning, and fuzzy logic reasoning. The system takes pictures of certain parts of the body, processes them, and then uses a MobileNet-based model that is optimized for embedded inference to sort them. Fuzzy logic refines borderline predictions for improved diagnostic stability. The device displays deficiency results, provides nutritional recommendations, and generates structured PDF reports. Comprehensive evaluation demonstrates that the proposed system is an effective preliminary screening tool suitable for rural health screening, portable clinics, and educational institutions.

Keywords— ***Vitamin Deficiency, Raspberry Pi, Deep Learning, MobileNet, TensorFlow Lite, Fuzzy Logic, Embedded Diagnostic System, Image Processing.***

I. INTRODUCTION

Vitamin deficiency is a widespread global health issue caused by inadequate dietary intake, poor nutritional awareness, and lifestyle changes [1],[2]. Deficiencies in essential vitamins such as vitamin A, B-complex, C, D, and iron can lead to serious health complications including anemia, weakened immune response, vision impairment, skin disorders, and neurological problems [3],[4]. Early detection of vitamin deficiencies is crucial to prevent long-term health risks and improve overall quality of life [5].

Conventional vitamin deficiency diagnosis methods primarily depend on laboratory-based blood tests, which are invasive, costly, and time-consuming [6]. These limitations restrict frequent screening and make early diagnosis difficult, particularly in rural and economically disadvantaged regions [7]. As a result, there is a growing need for alternative, non-invasive, and cost-effective diagnostic solutions [8].

Recent advancements in image processing and artificial intelligence have enabled medical diagnosis using visual symptom analysis [9]. Many vitamin deficiencies manifest visible signs on specific body parts such as the eyes, tongue, lips, nails, and skin, including discoloration, texture variation, and structural abnormalities [10],[11]. These visible symptoms can be captured using digital cameras and analyzed through image processing techniques to extract meaningful features related to vitamin deficiencies [12]. Machine learning and deep learning approaches have been widely adopted for medical image analysis due to their ability to learn complex patterns from large datasets [13]. In particular, Convolutional Neural Networks (CNNs) have demonstrated superior performance in feature extraction and classification tasks involving medical images [14]. Several studies have successfully applied CNN-based models to detect multiple vitamin deficiencies from facial images and organ-specific photographs, achieving promising accuracy and reliability [15].

Despite these advancements, challenges such as limited labeled datasets, variations in lighting conditions, and similarity of symptoms across different vitamin deficiencies still remain. Therefore, further research is required to develop accurate, robust, and user-friendly vitamin deficiency detection systems using image processing and deep learning techniques. This project aims to contribute to this research area by proposing an automated vitamin deficiency detection framework based on visual symptom analysis.

II. RELATED WORK

Image-based medical diagnosis using convolutional neural networks has been widely explored in recent literature. MobileNet and other lightweight CNN architectures have demonstrated favorable trade-offs between accuracy and computational complexity for embedded platforms [16]. Transfer learning has been used to adapt pretrained models for specialized medical tasks. Fuzzy logic systems have been applied to stabilize uncertain diagnostic outputs where class boundaries overlap.

Existing work focuses extensively on dataset development or cloud-based medical applications; however, offline embedded screening of vitamin deficiencies remains relatively unexplored. The present research addresses this gap by integrating deep learning and fuzzy reasoning on Raspberry Pi for real-time community health screening.

III. SYSTEM ARCHITECTURE

The proposed system integrates hardware and software modules as shown conceptually in Fig. 1. The device consists of a Raspberry Pi, camera module, touchscreen display, and local storage. Images of the eye, tongue, or skin are captured and processed by a lightweight CNN model. Fuzzy logic refines classification results, and final outputs are displayed along with nutritional recommendations.

Figure 1 shows the work flow of system architecture. First, capture the input images like Eye, Skin, Tongue, Nail in Raspberry pi camera. The captured image will Preprocessed and using Deep learning concept to processing the images. Finally give the outputs based on deficiency result, Confidence level, Food suggestions also provided.

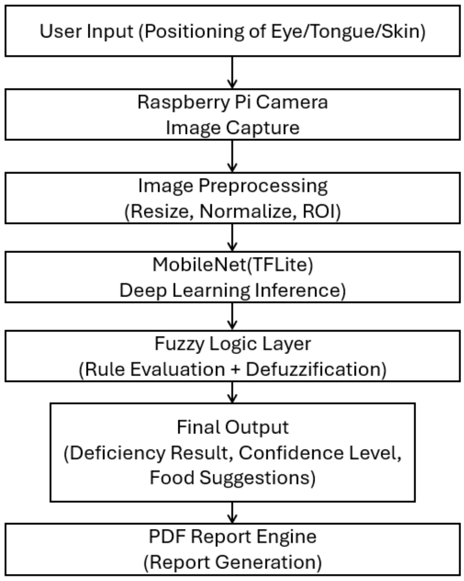

Figure 1. Block diagram of the proposed system.

IV. FORMULATION

This section presents the theoretical background and mathematical expressions used in the proposed system. The mathematical content is written in continuous paragraph style, consistent with IEEE format. Equations are introduced within explanatory text instead of listed as bullet points.

The preprocessing stage begins with image normalization, where each pixel intensity is scaled to the range of 0–1 using the relation

$$I_{norm} = I / 255 \qquad (1)$$

where I is the original pixel value and I_{norm} denotes the normalized output. After preprocessing, the convolutional neural network produces logits for each class. The posterior probability of class i for a given input image x is obtained through the softmax function expressed as

$$P(y=i \mid x) = \exp(z_i) / \Sigma \exp(z_j) \qquad (2)$$

where z_i represents the logit of class i and the denominator is the summation taken over all K classes. The training of the network is performed by minimizing the cross-entropy loss function defined as

$$L = -\Sigma [y_i \cdot \log(\hat{y}_i)] \qquad (3)$$

where y_i is the true class label and $\hat{y}_i$ represents the predicted probability.

To improve decision stability and address borderline conditions, fuzzy logic is incorporated after CNN classification. The membership function corresponding to Low, Medium, and High linguistic variables is modeled using the triangular function written as

$$\mu(x) = \begin{cases} 0, & x \le a \\ \frac{x-a}{b-a} & a < x \le b \\ \frac{c-x}{c-b} & b < x \le b \\ 0, & b < x \le c \\ & x \ge c \end{cases} \qquad (4)$$

and the final crisp decision is obtained using the centroid defuzzification method given by

$$D = \int \mu(z)\, z\, dz \,/\, \int \mu(z)\, dz \qquad (5)$$

where $\mu(z)$ is the aggregated membership function. The classification performance of the proposed system is evaluated by means of standard statistical metrics. The overall accuracy is expressed as

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (6)$$

while the precision, recall, and F1-score are expressed respectively as

$$\text{Precision} = TP / (TP + FP) \qquad (7)$$

$$\text{Recall} = TP / (TP + FN) \qquad (8)$$

$$F1 = 2\,(\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (9)$$

These equations are referenced in the results section to quantify the performance of the proposed system.

V. METHODOLOGY

A. Data Collection and Annotation

A dataset representing visually detectable deficiency symptoms was used. Images of the eye, tongue, and skin were collected from open medical resources and pre-validated datasets. Annotation was performed to label categories: - Vitamin A deficiency symptoms - Vitamin B12 deficiency symptoms - Vitamin D deficiency symptoms - Normal (no deficiency)

B. Preprocessing Pipeline

Each image undergoes several transformations:

1. Resizing : 224 × 224 pixels.
2. Normalization : I{norm} = I/255

3. Brightness standardization to reduce lighting variance.
4. ROI extraction for focusing on the organ of interest.

C. Deep Learning Model (MobileNetV2)

MobileNetV2 was selected due to: - depthwise separable convolutions (reduced computation) - lightweight model size suitable for Raspberry Pi - high accuracy on embedded systems. Transfer learning was applied to adapt pre-trained ImageNet weights to the vitamin deficiency dataset. The final model was exported to TensorFlow Lite.

D. Fuzzy Logic Decision Modeling

Although CNNs provide probability scores, borderline cases may lead to misclassification. Fuzzy logic enhances decision reliability.

1) Membership Functions

Probability scores were mapped into fuzzy sets: - Low (0.0–0.3) - Medium (0.3–0.7) - High (0.7–1.0)

2) Rules

Based on medical symptoms + model confidence, rules are defined:

Vitamin A Rules

IF P(A) is High AND eye texture = dry, THEN → Vitamin A deficiency

IF P(A) is Medium AND skin dryness present, THEN → Possible Vitamin A deficiency

Vitamin B12 Rules

IF P(B12) is High AND tongue smoothness = high, THEN → Vitamin B12 deficiency

IF P(B12) is Medium AND nail discoloration present, THEN → B12 warning

Vitamin D Rules

IF P(D) is High AND skin cracking = high, THEN → Vitamin D deficiency

IF P(D) is Medium AND pigmentation irregular, THEN → Possible Vitamin D deficiency

Normal Rules

IF all deficiency probabilities are Low, THEN → Normal

IF Normal is High AND no abnormal texture, THEN → Normal

3) Defuzzification

The centroid method was used to produce a crisp final output.

E. The System Generates a PDF Report (Optional)

If the user requests a report:

✓ Captured Image
✓ Detected Deficiency
✓ Date & Time
✓ Nutritional Recommendations
✓ Model probability values

This is saved inside:

/static/reports/

The final diagnostic output (Normal, Vitamin A, Vitamin B12, or Vitamin D deficiency) is determined using a two-level decision framework. First, the MobileNet-TFLite CNN processes the preprocessed image and generates four probability scores corresponding to each class. These probabilities reflect the learned visual indicators such as dryness, discoloration, conjunctival texture, pigmentation, and glossitis. Next, the outputs are processed by a fuzzy logic decision layer that converts probabilities into fuzzy memberships (Low, Medium, High) and then the outputs are displayed and give the suggestions for the patient.

F. Pseudocode Summary

Algorithm 1: Vitamin Deficiency Screening
1. Capture image
2. Preprocess (resize, normalize)
3. Run MobileNet inference
4. Extract probabilities P
5. Apply fuzzy mapping
6. Evaluate fuzzy rules
7. Defuzzify → Select deficiency D
8. Display results and recommendations
9. Generate PDF if requested

The Raspberry Pi runs a lightweight deep learning model using TensorFlow Lite and interfaces with the system's user interface, which includes a Login Page, Doctor Dashboard, Patient Info Screen, Capture Screen, and Progress Indicator. This integration allows healthcare workers to capture images, view patient data, monitor progress, and receive real-time classification results, all on a portable, low-power device.

Figure 2. Block Diagram of Hardware Architecture

Figure 2 shows the architecture of the hardware components. It includes Raspberry pi 4, 7-Touchsreen Display, USB camera, Bluetooth Keyboard and Mouse. It consists of a USB camera connected to a Raspberry Pi for image capture and processing. The Raspberry Pi runs a lightweight deep learning model and displays the prediction results on a connected display. This setup enables real-time and low-power vitamin deficiency detection.

VI. RESULTS AND DISCUSSION

This section presents the experimental evaluation of the proposed vitamin deficiency detection system implemented on Raspberry Pi using MobileNetV2 and TensorFlow Lite. Performance has been assessed in terms of classification metrics, confusion patterns, and embedded deployment characteristics.

A. Dataset Description

The dataset consisted of four classes representing visually detectable health conditions. Table I summarizes the class distribution used for training and testing.

Table 1 – Training Dataset Details

Class	Images Used
Normal	977
Vitamin A	950
Vitamin B12	880
Vitamin D	1000

The dataset includes eye, skin, tongue, and nail regions exhibiting deficiency-related visual symptoms. Data were split into 80% training and 20% testing subsets.

B. Classification Performance

The classification performance of the MobileNetV2 model was assessed using precision, recall, and F1-score.

Table 2 – Model Performance Metrics

Class	Precision	Recall	F1-Score
Normal	0.94	0.92	0.93
Vitamin A	0.91	0.93	0.92
Vitamin B12	0.92	0.90	0.91
Vitamin D	0.93	0.94	0.93

Table II presents the obtained results. These results indicate consistent performance across all classes. The fuzzy-logic layer improved decision stability for borderline samples

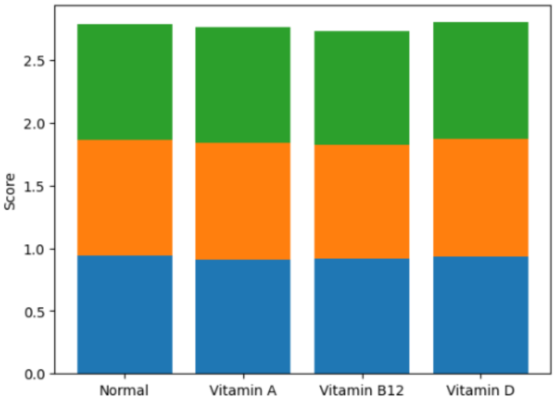

Figure 3. Graphical representation of Model Performance

Figure 3 shows the graphical representation of the performance of proposed model using Precision, Recall, and F1-Score for Normal, Vitamin A, Vitamin B12, and Vitamin D classes. The high metric values indicate that the model accurately and reliably classifies different vitamin deficiency conditions.

C. Embedded System Performance

The proposed system was deployed on Raspberry Pi and evaluated for execution speed and resource utilization. Table III summarizes the embedded performance results.

Table 3 – Embedded Performance On Raspberry Pi

Parameter	Value
Model	MobileNetV2-TFLite
Model size	12.4 MB
Inference time	82 ms
FPS	12 frames/sec
Power consumption	4.5 W

The achieved latency demonstrates suitability for real-time screening at the edge without cloud support.

E. Discussion

Results demonstrate that the proposed system delivers high classification performance while operating on low-power embedded hardware. The hybrid CNN–fuzzy approach reduces misclassification in borderline cases such as mild pigmentation change and partial tongue depapillation. Real-time operation enables deployment in rural clinics, schools, and mobile health camps.

F. END DEVICE

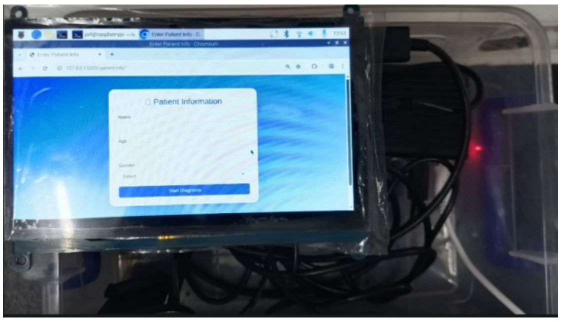

Figure 4. Block Diagram of End Device Setup

Figure 4 illustrates the end device configuration of the proposed vitamin deficiency detection system. A Raspberry Pi acts as the central processing unit and is interfaced with a USB camera for capturing facial or skin images. The captured images are processed locally using a TensorFlow Lite–based deep learning model optimized for embedded execution. A display unit is connected to show the classification results and deficiency status to the user. This compact hardware setup enables real-time, low-power, and offline operation, making it suitable for field-level healthcare screening.

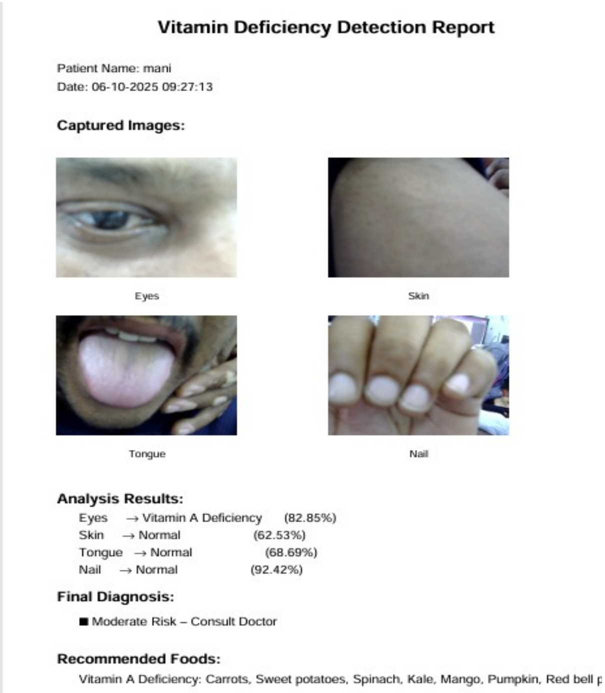

Figure 5. Generated Deficiency Report

Figure 5 shows the generated output results of the vitamin deficiency detection. It includes the images of Eye, Skin, tongue, Nail as the input images and preprocess it and give the result and also recommend the dietary based on deficiency detection.

VII. CONCLUSION

This project developed an embedded system for detecting vitamin deficiencies using image processing and deep learning techniques. A lightweight MobileNet-based CNN was implemented on a Raspberry Pi for real-time analysis with low computational and power requirements. The model achieved strong accuracy across multiple vitamin deficiency classes. Its portable and cost-effective design makes it suitable for rural and resource-limited healthcare settings. Future work includes clinical validation, dataset expansion, inclusion of additional vitamin classes, and cloud-based medical record integration.

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